

Alpha-Beta Pruning Algorithm

PROGRAM:

```
def alphabeta(depth, node_index, is_max, values, max_depth, alpha, beta):

    if depth == max_depth:
        return values[node_index]

    if is_max:
        best = float('-inf')
        for i in range(2):
            val = alphabeta(depth + 1, node_index * 2 + i,
                            False, values, max_depth, alpha, beta)
            best = max(best, val)
            alpha = max(alpha, best)

            if beta <= alpha:
                break
        return best
    else:
        best = float('inf')
        for i in range(2):
            val = alphabeta(depth + 1, node_index * 2 + i,
                            True, values, max_depth, alpha, beta)
            best = min(best, val)
            beta = min(beta, best)

            if beta <= alpha:
                break
        return best

values = [3, 5, 6, 9, 1, 2, 0, -1]
tree_depth = 3
result = alphabeta(0, 0, True, values, tree_depth,
                   float('-inf'), float('inf'))
print("Optimal value using Alpha-Beta:", result)
```

OUTPUT:

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\kames\OneDrive\Documents\alpha-beta.py =====
Optimal value using Alpha-Beta: 5
>>>
```